Lab 1. Setting Up the Course Lab Environment

Overview

Before we dive into the hands-on activities, let's configure the lab environment with the following tools:

- **Docker:** Used by kind to provision the Kubernetes cluster.
- **kind:** Used to create Kubernetes clusters.
- **kubectl:** Used to manage the Kubernetes cluster.
- **Helm:** Used for Kubernetes package management.
- **Siege:** A load testing tool.

Exercise 1.1: Set Up the Course Lab Environment

Linux Environment Setup

While we present various setup options in this section, we will use an Ubuntu 24.04 host and a Kubernetes cluster setup using `kind`.

To set up **Docker**, follow these steps.

1. Install Docker:

```
curl -fsSL https://get.docker.com/ | sh
```

2. Enable Docker to start on boot:

```
sudo systemctl enable --now docker
```

3. Check that the Docker service is running:

```
sudo systemctl status docker
```

4. Add current user to Docker group:

```
sudo usermod -aG docker $USER
```

5. Refresh shell session **(by exiting & logging in again)**.
6. Verify Docker installation:

```
docker ps
```

To install the latest version of **kubectl**, execute the commands below.

1. Download the binary:

```
curl -sSL -O "https://dl.k8s.io/release/$(curl -L -s
https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl"
```

2. Modify permissions:

```
chmod +x kubectl
```

3. Move to `/usr/local/bin`:

```
sudo mv kubectl /usr/local/bin
```

To install the latest **Helm** version, perform the following steps.

1. Download Helm (if the link doesn't work, update the link with a version from [GitHub releases]()):

```
curl -sSL -O https://get.helm.sh/helm-v3.19.0-linux-amd64.tar.gz
```

2. Extract the downloaded archive:

```
tar -zxf helm-v3.19.0-linux-amd64.tar.gz
```

3. Move the Helm binary to `/usr/local/bin`:

```
sudo mv linux-amd64/helm /usr/local/bin/helm
```

To install the latest version of **Siege,** execute the following command.

1. Install the binary:

```
sudo apt-get install siege
```

Siege is a multi-threaded HTTP/FTP load tester and benchmarking utility.

To install **`kind`**, run the following command.

1. For AMD64 / x86_64:

```
[ $(uname -m) = x86_64 ] && curl -Lo ./kind
https://kind.sigs.k8s.io/dl/v0.30.0/kind-linux-amd64
```

2. For ARM64:

```
[ $(uname -m) = aarch64 ] && curl -Lo ./kind
https://kind.sigs.k8s.io/dl/v0.30.0/kind-linux-arm64
```

3. Modify permissions:

```
chmod +x ./kind
```

4. Move the kind binary `/usr/local/bin`:

```
sudo mv ./kind /usr/local/bin/kind
```

5. Create cluster:

```
kind create cluster
```

```
Creating cluster "kind" ...
 ✓ Ensuring node image (kindest/node:v1.34.0) 🖼
 ✓ Preparing nodes 📦
 ✓ Writing configuration 📜
 ✓ Starting control-plane 🕹️
 ✓ Installing CNI 🔌
 ✓ Installing StorageClass 💾
Set kubectl context to "kind-kind"
You can now use your cluster with:

kubectl cluster-info --context kind-kind

Have a nice day! 👋
```

6. Verify cluster:

```
kubectl get ns
```

```
NAME                 STATUS   AGE
default              Active   36s
kube-node-lease      Active   36s
kube-public          Active   36s
kube-system          Active   36s
local-path-storage   Active   30s
```

Metric Server Setup in Kubernetes

To install metrics server in Kubernetes, execute the commands below.

1. Add Helm repository:

```
helm repo add metrics-server https://kubernetes-sigs.github.io/metrics-server/
```

`metrics-server` has been added to your repositories.

2. Install metrics server:

```
helm upgrade --install metrics-server metrics-server/metrics-server -n
kube-system --set args[0]=--kubelet-insecure-tls
```

```
NAME: metrics-server
LAST DEPLOYED: Wed Oct 15 16:20:19 2025
NAMESPACE: kube-system
STATUS: deployed
REVISION: 1
TEST SUITE: None
NOTES:
***********************************************************************
* Metrics Server                                                      *
***********************************************************************
  Chart version: 3.13.0
  App version:   0.8.0
  Image tag:     registry.k8s.io/metrics-server/metrics-server:v0.8.0
***********************************************************************
```

3. Verify installation:

```
kubectl get pods -n kube-system -l=app.kubernetes.io/name=metrics-server
```

```
NAME                              READY   STATUS    RESTARTS   AGE
metrics-server-55fd7d4789-v6dpw   1/1     Running   0          63s
```

Congratulations, your environment is ready!